LE, Bach

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Google Scholar: 1622 citations, h-index: 15, i10-index: 17
https://scholar.google.com/citations?user=AUJWzE8AAAAJ&hl=en

"Curiosity is the wick in the candle of learning" – William Arthur Ward

Summary

I am interested in Software Engineering (SE), particularly Empirical Software Engineering (Human/User Studies), AI for SE and SE for AI, and Program Analysis. I built systems that involve mining massive amounts of past data, e.g., code bug fixes, for actionable insights to inform future software development. My research has attracted **~$500K AUD as Sole PI** from the Australian Research Council. I am also a collaborator of more than **$2M AUD** of research funding awarded by NSF (USA), and Singapore Ministry of Education. My work has inspired the first-ever automated bug fixing system, namely GetAFix, deployed at Meta in 2019.

I am currently managing a ~$500K AUD of funding from the Australian Research Council Discovery Award (ARC DECRA) as sole PI, supervising three PhD students, have graduated three master's students by research, and led 40 master's students (~10 students per group) working with industry partners to deliver various successful projects such as web design, blockchain-based payment system, etc. Under my primary supervision, my students have won prestigious prizes such as **Distinguished Paper Award**, **Google PhD Fellowship**, **University Endeavor Awards, and Best Undergraduate Thesis Award**.

I lecture on various subjects at The University of Melbourne such as Models of Computation (600 students), Programming Language and Implementation – Compiler Theory (45 students), Object-oriented Software Development (400 students), etc.

Appointments

Assistant Professor | The University of Melbourne, Australia | July'19 – Present, officially Tenured/Confirmed from September 2023.

ARC DECRA Fellow – Sole PI of $488K AUD awarded by the Australian Research Council for Discovery Early Career Researcher| January 2022 – 2024

Education

PostDoc | Carnegie Mellon University, Silicon Valley, USA | June'18 – June'19

- *Project*: ISSTAC: Integrated Symbolic Execution for Space-Time Analysis of Code, aiming at software vulnerabilities detection/prevention, funded by DARPA for Space-Time Analysis of Cyber-Security.
- *Advisor*: ACM Distinguished Scientist & Associate Prof. Corina Pasareanu

Ph.D. Residency Training | Carnegie Mellon University, Pittsburgh, USA | Aug'15 – June'16

- *Project*: Automatic software repair, as part of research collaboration program LARC-CMU of Singapore Management University with Carnegie Mellon University as a strategic partner.
- *Advisor*: Associate Prof. Claire Le Goues

Ph.D. Candidate | Singapore Management University, Singapore | Jan'14 – May'18

- *Thesis*: "Overfitting in Automated Program Repair: Challenges and Solutions". Automatic software repair using Machine Learning, Data Mining, Program Synthesis, Analysis and Verification.
- *Advisor*: Prof. David Lo

B. Tech. | Hanoi University of Science and Technology, Vietnam | Aug'07 – May'12

- *Thesis*: HIP/SLEEK: Static verification of heap-manipulating programs via Separation Logic.
- Degree obtained under collaboration program with Keio University, Japan, supported by Japanese ODA
- *Advisors*: Associate Prof. Quyet-Thang Huynh and Associate Prof. Wei-Ngan Chin

Additional Research Experience

Research Assistant | National University of Singapore, Singapore | Feb'12 – May'12 & Aug'12 – Nov'13

- *Project*: HIP/SLEEK: Static verification of heap-manipulating programs via Separation Logic, aiming at verifying C-like programs using SAT, Separation Logic, and Hoare-style deductive verification.
- *Advisor*: Associate Prof. Wei-Ngan Chin

Research Intern | Nagoya University, Japan | Aug'11

- *Project*: Building a Twitter-like social network using C#.
- *Advisor*: Prof. Toyohide Watanabe

Industry Internship

FSoft (FPT Software) | Vietnam | Aug'11

- *Project*: Building a web service application using JSP, as part of undergraduate internship program collaborated between Hanoi University of Science and Technology, and FPT Software company.
- *Company website*: https://www.fpt-software.com/

Certificates

- Fundamental Information Technology Engineer Skill Standards (FE) Certification, **Japan IT Engineers Examination Center**. Certified by the Ministry of Science and Technology, 2011

Honors & Awards

- **Distinguished Paper Award** at The 39th International Conference on Software Maintenance and Evolution (ICSME), 2023.
- PhD Scholarship at Singapore Management University, 2014 – 2018.
- PhD Exchange Scholarship to Carnegie Mellon University, 2015 – 2016.
- Best Undergraduate Thesis, Higher Education Development Support Project on ICT (HEDSPI), Hanoi University of Science and Technology (HUST), 2012.
- Undergraduate Internship Scholarship to Nagoya University, 2011.

- First Prize in Mathematics (best among ~200 selected students), Nghe An Province, Vietnam, 2005.
- First Prize in Mathematics (best among ~200 selected students), Nghe An Province, Vietnam, 2000.

Languages

- **English**: Professional working level
- **Japanese**: Intermediate level (JLPT Level 2, obtained in 2009)
- **Vietnamese**: Native

Computer Skills

- **Programming**: Scala, Java, Ocaml, Bash, Python, Haskell, Latex, C, C#, C++, Prolog
- **Machine learning**: Classification (WEKA library), Active Learning (JCLAL library)
- **Databases**: PostgreSQL, MYSQL, Google BigQuery
- **Eclipse Plugin**: See https://xuanbachle.github.io/semanticsrepair/

Note: I developed several projects using the above programming languages during my PhD study. The largest project that I developed involves a mix of 14K lines of code in Scala and 3K lines of code in Java.

Articles

Note: My students are indicated by double underline.

[31] **Are We Ready to Embrace Generative AI for Software Q&A?**
Bowen Xu, Thanh-Dat Nguyen, Thanh Le-Cong, Thong Hoang, Jiakun Liu, Kisub Kim, Chen GONG, Changan Niu, Chenyu Wang, Xuan-Bach D. Le, David Lo
The 38th IEEE/ACM International Conference on Automated Software Engineering **(ASE)**, New Ideas Track, 2023.

[30] **The Future Can't Help Fix The Past: Assessing Program Repair In The Wild.**
Vinay Kabadi, Dezhen Kong, Siyu Xie, G. Arthar, Lingfeng Bao, Tien-Duy BL, Xuan-Bach D. Le, and David Lo
The 39th IEEE International Conference on Software Maintenance and Evolution **(ICSME)**, 2023.
Won **Distinguished Paper Award** – the only paper among the accepted paper that won the prize. (Acceptance rate ~22%)

[29] **Multi-Granularity Detector for Vulnerability Fixes.**
Truong Giang Nguyen, Thanh Le-Cong, Hong Jin Kang, Ratnadira Widyasari, Chengran Yang, Zhipeng Zhao, Bowen Xu, Jiayuan Zhou, Xin Xia, Ahmed Hassan, Xuan-Bach D. Le, and David Lo
Transactions on Software Engineering **(TSE)** Journal, 2023

[28] **Invalidator: Automated Patch Validation via Dynamic Invariant Inference**.
Cong-Thanh Le, Duc-Minh Luong, Xuan-Bach D. Le, David Lo, Nhat-Hoa Tran, and Quyet-Thang Huynh.
Transactions on Software Engineering **(TSE)** Journal, 2023.

[27] **Chronos: Time-Aware Zero-Shot Identification of Libraries from Vulnerability Reports.**
Y. Lyu, Thanh Le-Cong, Hong Jin Kang, R. Widyasari, Z. Zhao, Xuan-Bach D. Le, Ming Li, David Lo
The 45th IEEE/ACM International Conference on Software Engineering (**ICSE**), Research Track, 2023.

[26] **An Efficient Cyclic Entailment Procedure in a Fragment of Separation Logic.**

Quang-Loc Le, Xuan-Bach D. Le
The 26th International Conference on Foundations of Software Science and Computation Structures (**FoSSaCS**), Research Track, 2023.

[25] **VulCurator: A Vulnerability-Fixing Commit Detector.**
Truong Giang Nguyen, Thanh Le-Cong, Hong Jin Kang, Xuan-Bach D. Le, David Lo
The 21st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**ESEC/FSE**), Demo Track, 2022.

[24] **AutoPruner: Tranformer-based Call Graph Pruning.**
Thanh Le-Cong, Hong Jin Kang, Truong Giang Nguyen, S. Haryono, David Lo, Xuan-Bach D. Le, and Quyet-Thang Huynh
The 21st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**ESEC/FSE**), Research Track, 2022.

[23] **FFL: Fine grained Fault Localization for Student Programs via Syntactic and Semantic Reasoning.**
Thanh-Dat Nguyen, Thanh Le-Cong, Duc-Minh Luong, Van-Hai Duong, Xuan-Bach D. Le, David Lo and Quyet-Thang Huynh
The 38th IEEE International Conference on Software Maintenance and Evolution (**ICSME**), Research Track, 2022.

[22] **Test Mimicry to Assess the Exploitability of Library.**
Hong Jin Kang, Truong Giang Nguyen, Xuan-Bach D. Le, David Lo, Corina Pasareanu
The 31st International Symposium on Software Testing and Analysis (**ISSTA**), Research Track, 2022.

[21] **Toward the Analysis of Graph Neural Networks.**
Thanh-Dat Nguyen, Thanh Le-Cong, ThanhVu H Nguyen, Xuan-Bach D. Le, Quyet-Thang Huynh
The 44th International Conference on Software Engineering (**ICSE**), NIER Track, 2022.

[20] **REFIXAR: Multi-version Reasoning for Automated Repair of Regression Errors.**
Xuan-Bach D. Le, Quang-Loc Le
The 32nd International Symposium on Software Reliability Engineering (**ISSRE**), Research Track, 2021.

[19] **Usability and Aesthetics: Better Together for Automated Repair of Web Pages.**
Cong-Thanh Le, Xuan-Bach D. Le, Quyet-Thang Huynh, Phi Le Nguyen
The 32nd International Symposium on Software Reliability Engineering (**ISSRE**), Research Track, 2021.

[18] **More Reliable Test Suites for Dynamic Program Repair by Using Counterexamples.**
Amirfarhad Nilizadeh, Marlon Calvo, Gary T. Leavens, Corina Pasareanu and Xuan-Bach D. Le
The 32nd International Symposium on Software Reliability Engineering (**ISSRE**), Research Track, 2021.

[17] **Exploring True Test Overfitting in Dynamic Program Repair using Formal Methods.**
Amirfarhad Nilizadeh, Gary T. Leavens, Xuan-Bach D. Le, Corina Pasareanu, and David R. Cok
International Conference on Software Testing Verification and Validation (**ICST**), Research Track, 2021

[16] **Smart Contract Development: Challenges and Opportunities**.
Z. Weiqin, L. David, S. K. Pavneet, Xuan-Bach D. Le, F. Yang, Zhenyu Chen, and Baowen Xu
Transactions on Software Engineering ***(TSE)***, 2019

[15] **SAFFRON: Adaptive Grammar-based Fuzzing for Worst-Case Analysis**
Xuan-Bach D. Le, Corina Pasareanu, Rohan Padhye, David Lo, Willem Visser, and Koushik Sen
Java Pathfinder Workshop collocated with ASE ***(JPF)***, 2019.

[14] **Symbolic Pathfinder for SV-COMP**
Yannic Noller, Corina Pasareanu, Aymeric Fromherz, Xuan-Bach D. Le, and Willem Visser
Competition on Software Verification (**SV-COMP**), 2019, held at TACAS.

[13] **On Reliability of Patch Correctness Assessment**.
Xuan-Bach D. Le, Lingfeng Bao, David Lo, Xia Xin, Shanping Li, and Corina Pasareanu
In *International Conference on Software Engineering (**ICSE**),* Research Track, 2019

[12] **Overfitting in Semantics-based Automated Program Repair.**
Xuan Bach D Le, David Lo, Ferdian Thung, and Claire Le Goues.
In *International Conference on Software Engineering (**ICSE**), Journal First Track, 2018.*
In *Empirical Software Engineering (**EMSE**) Journal*, 2018.

[11] **S3: Syntax- and Semantics-guided Repair Synthesis via Programming by Examples.**
Xuan Bach D Le, David Lo, Duc-Hiep Chu, Claire Le Goues, and Willem Visser.
In *European Software Engineering Conference and the ACM SIGSOFT Symposium on the Foundations of Software Engineering (**ESEC/FSE**)*, Research Track, 2017.

[10] **JFix: Semantics-based Repair of Java Programs via Symbolic Pathfinder.**
Xuan Bach D Le, Duc-Hiep Chu, David Lo, Claire Le Goues, and Willem Visser.
In *26th International Symposium on Software Testing and Analysis (**ISSTA**)*, Demo Track, 2017.

[9] **Xsearch: A Domain-specific Cross-language Relevant Question Retrieval Tool.**
Bowen Xu, Xing Zhen-chang, Xia Xin, David Lo, and Xuan-Bach D Le.
In *European Software Engineering Conference and the ACM SIGSOFT Symposium on the Foundations of Software Engineering (**ESEC/FSE**)*, Demo Track, 2017.

[8] **Towards Efficient and Effective Automatic Program Repair.**
Xuan-Bach D Le.
In *Proceedings of the 31st International Conference on Automated Software Engineering (**ASE**)*, Doctoral Symposium, 2016.

[7] **Enhancing Automated Program Repair with Deductive Verification**.
Xuan-Bach D Le, Quang Loc Le, David Lo, and Claire Le Goues.
In *32nd International Conference on Software Maintenance and Evolution (**ICSME**)*, ERA Track, 2016.

[6] **Empirical Study on Synthesis Engines for Semantics-based Program Repair.**
Xuan Bach D Le, David Lo, and Claire Le Goues.
In *32nd International Conference on Software Maintenance and Evolution (**ICSME**)*, ERA Track, 2016.

[5] **Recommending Code Changes for Automatic Backporting of Linux Device Drivers.**
Ferdian Thung, Xuan-Bach D Le, David Lo, and Julia Lawall.
In *32nd International Conference on Software Maintenance and Evolution* (***ICSME***), Research Track, 2016.

[4] **History Driven Program Repair.**
Xuan Bach D Le, David Lo, and Claire Le Goues.
In *23rd International Conference on Software Analysis, Evolution, and Reengineering* (***SANER***), Research

Track, 2016.

[3] **Should Fixing These Failures be Delegated to Automated Program Repair**?
Xuan-Bach D Le, Tien-Duy B Le, and David Lo.
In *26th International Symposium on Software Reliability Engineering* (***ISSRE***), Research Track, 2015.

[2] **Synergizing Specification Miners Through Model Fissions and Fusions**.
Tien-Duy B Le, Xuan-Bach D Le, David Lo, and Ivan Beschastnikh.
In *30th International Conference on Automated Software Engineering* (***ASE***), Research Track, 2015.

[1] **Active Semi-supervised Defect Categorization**.
Ferdian Thung, Xuan-Bach D Le, and David Lo.
In *23rd International Conference on Program Comprehension* (***ICPC***), Research Track, 2015.

PhD thesis: **Overfitting in automated program repair: Challenges and Solutions.**
Xuan Bach D Le
PhD Thesis, Singapore Management University, Singapore, May 2018.

Recent Unpublished Work

[32] **Adversarial Patch Generation for Automatic Program Repair,** Aziz Alhefdhi, Cong-Thanh Le, Khanh-Hoa Dam, under submission to Software Quality Journal.
[33] **Refining ChatGPT-Generated Code: Characterizing and Mitigating Code Quality Issues,** Yue Liu, Cong-Thanh Le, Ratnadira Widyasari, Chakkrit Tantithamthavorn, Li Li, Xuan-Bach D. Le, David Lo, under submission to TSE'23.
[34] **PatchZero: Zero-Shot Automatic Patch Correctness Assessment,** Xin Zhou, Bowen Xu, Kisub Kim, DongGyun Han, Cong-Thanh Le, Junda He, Xuan-Bach D. Le, David Lo under submission to TSE'23.
[35] **Mining Graphs to Attack Code Models,** Thanh-Dat Nguyen, Zhou Zhang, Xuan-Bach D. Le, Patanamon Thongtanunam, David Lo, under submission.
[36] **GNN-Infer: Property Inference for Graph Neural Network,** Thanh-Dat Nguyen, Hieu Nguyen, Cong-Thanh Le, Xuan-Bach D. Le, David Lo, ThanhVu Nguyen, Corina Pasareanu, under submission.

Student Supervision

PhD students:

- Vinay Kabadi, Part-time PhD student at The University of Melbourne. Thesis topic: empirical study on software debugging, January 2020 - now.
  - Award: **Distinguished Paper Award** at ICSME'23.
- Thanh-Dat Nguyen, Full-time PhD student at The University of Melbourne. Thesis topic: Program Synthesis for Automated Information Extraction, April 2021 - now.
- Cong-Thanh Le, Joining PhD program at The University of Melbourne in January 2023. Cong-Thanh has been under my supervision, graduated from Hanoi University of Science and Technology (HUST) in 2019.
  - Awards: **Best Undergraduate Thesis** from HUST in 2019. **Google PhD Fellowship** in 2023.

Master's students:

- Ishan Maholtra, master's student from The University of Melbourne, 2019 – 2020 (graduated). Topic: Automated Repair of student programming assignments.

- Chijiang Yang, master's student from The University of Melbourne, 2020-2021 (graduated). Topic: Unified debugging: how far are we?
- Shubham Parth, master's student from The University of Melbourne, 2020-2021 (graduated). Topic: Smart contract fuzzing.

Undergraduate students:

- Thanh Le Cong, undergraduate student at Hanoi University of Science and Technology, 2019 – now
  - Award: **Best Undergraduate Thesis Award**. Topic: Automated Program Repair
- Siyu Xie, female undergraduate student from Zhejiang University, China, visited SMU during 2017 - 2018. This resulted in a full research paper on automatic repair, now under submission.
  - Award: **Distinguished Paper Award** at ICSME'23 (together with Vinay Kabadi)
- Hanliang Zhang, undergraduate student from Peking University, China during Google Summer of Code, 2019

Project Supervision

I regularly supervise groups of ~10 students per group working on their master's projects across organizations, e.g., university and industry. The projects span one semester (12 weeks) or three semesters (1 year). Notable projects, **that won University Endeavor Award**, include:

- **Reusable coffee cup**: build an online management system that allows users to borrow and return coffee cups when buying coffee. (1-year project, 2019 - 2020)
- **Apollo Music**: build a blockchain-based system that allows the music industry to manage payments for music shows, e.g., online blockchain-based smart contract to pay artists and promoters for music shows. (1-year project, 2020 - present)

Teaching

- **Lecturer/Coordinator**: Models of Computation, The University of Melbourne, 2022 (~600 students enrolled). This is a core (required) subject in Computer Science at Unimelb.
- **Lecturer/Coordinator**: Models of Computation, The University of Melbourne, 2021 (~600 students enrolled). This is a core (required) subject in Computer Science at Unimelb.
- **Lecturer**: Programming Language Implementation (Compiler Theory), The University of Melbourne, 2020 (~40 students enrolled) – Graduate Level
- **Lecturer/Coordinator**: Object-Oriented Software Development, The University of Melbourne, 2019, 2021, 2023 (~350 students enrolled)
- **Lecturer/Coordinator**: Masters Software Engineering Project, The University of Melbourne, 2020 (~120 students enrolled) – Graduate Level
- **Guest Lecturer**: Program Analysis and Modeling, Graduate level, SMU, 2018
- **Teaching Assistant**: Software Mining and Analysis, Graduate level, SMU, 2018
- **Teaching Assistant**: Analytics Foundations, Undergraduate level course, SMU, 2017

Activities

- **Invited talks:**
  - Invited speaker at NASA Robust Software Engineering, California, USA. Topic: Recent Advances on Automated Program Repair.

- Top 4 speakers invited to serve on the expert panel to talk about the Future of Automated Program Repair at Automated Program Repair workshop, collocated with ICSE'23 in Melbourne, Australia.
- Invited speaker at Monash University, 2019 and 2023. Topic: Automated Software Debugging and Repair.

- **Charing role**:
  - Chair of Software Engineering Track for the 11th International Symposium on Information and Communication Technology (SoICT), 2022
  - Chair of Demonstration Track for the IEEE International Conference on Blockchain and Cryptocurrency (ICBC), 2022
  - Chair of Student Research Competition for the 36th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2021
- **Committee Member** for:
  - The 45th International Conference on Software Engineering (ICSE), 2023, Research Track
  - The 44th International Conference on Software Engineering (ICSE), 2022, Software Engineering in Practice Track
  - The 43rd International Conference on Software Engineering (ICSE), 2021, New Ideas Track
  - The 36th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2021. Research Track
  - The ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2020, Research Track
  - The 35th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2020. Research Track
  - The 42nd International Conference on Software Engineering (ICSE), 2020, New Ideas Track
  - The ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), Artifact Evaluation Committee, 2019
  - The 19th IEEE International Conference on Software Quality, Reliability, and Security (QRS), 2019
  - International Workshop on Intelligent Bug Fixing (IBF), 2019
- **Reviewer** for:
  - IEEE Transactions on Network and Service Management (TNSM), 2022
  - Transactions on Software Engineering (TSE), 2019, 2020, 2021, 2022
  - Transactions on Software Engineering and Methodology (TOSEM), 2018, 2019, 2020, 2021
  - Software Testing, Verification and Reliability (STVR), 2018
  - Information and Software Technology (IST), 2018
  - Journal of Computer and Communications (JCC), 2018
  - Journal of Systems and Software (JSS), 2018
  - Empirical Software Engineering Journal (EMSE), 2017, 2021
- **Sub-reviewer** for: *2013*: SAC; *2014*: POPL; *2016*: APSEC, ICECCS, MSR (data track), SATE, SAC; *2017*: IWESEP, ICSME, VL/HCC, RV; *2018*: SCAM, ESEC/FSE (Demo Track)
- **Student volunteer**: ASE 2016, Singapore.
- **Oral conference presentations**:
  - ESEC/FSE 2017, Paderborn, Germany.
  - ASE 2016, Singapore.
  - ISSRE 2015, Gaithersburg, USA.

Software

- JFix and S3: Semantics-based repair of Java programs (ISSTA'17 and FSE'17), built atop Symbolic PathFinder – a notable symbolic execution engine for Java.
  - https://xuanbachle.github.io/semanticsrepair/
- HDRepair: Search-based program repair (SANER'16) using data mining and genetic programming, built atop of gSpan graph miner and Watchmaker evolutionary framework.
  - https://github.com/xuanbachle/bugfixes

Grants

- External grants:
  - ARC DECRA 2022 (Sole PI, $488K AUD): "*Practical Automated Software Bug Fixing via Syntactic and Semantic Analyses*", $438K AUD awarded by ARC and $50K AUD awarded by The University of Melbourne.
- Internal grants:
  - Grant for hiring a postdoc position starting from January 2024 to December 2024, awarded by The University of Melbourne in recognition of my DECRA'22.
  - Seed grant: "*Formally Verified Syntax- and Semantics-guided Automatic Proof Repair*", with Christine Rizkallah, $8K AUD, awarded by Computer Science group, The University of Melbourne.
  - Sole PI, $20K AUD, "*Refixar: Practical Automated Repair of Software Regression Errors*", awarded by The University of Melbourne.
- Collaborator of research grants (awarded to collaborators, not The University of Melbourne):
  - Collaborator and assisted in writing part of funding proposal: *Uncovering vulnerabilities in Machine Learning Frameworks via Software Composition Analysis and Directed Grammar-based Fuzzing*, awarded $500K SGD by Singapore government. This project is a collaboration between Singapore Management University (PI), Veracode (security company in Singapore, PI), University of Melbourne (collaborator), and Carnegie Mellon University (collaborator).
  - Assisted in writing part of funding proposal: *Human-Guided hybrid Search* (HUGS), a joint proposal between Carnegie Mellon University, University of California Santa Barbara, University of California Berkeley, and University of Wisconsin Madison, awarded NSF Medium $1M USD. I contributed to writing semantics-based repair using symbolic execution and program synthesis.

References

Available upon request.